

Laurent Series of rational polynomial.

Let $f(z)$ and $g(z)$ be polynomials over $\mathbb{C}$, relatively prime

Consider the fraction $\frac{g(z)}{f(z)}$.

[Silverman, Exercise in Sec 9.1.

For $0 < |a| < |b|$, expand $f(z) = \frac{1}{[z-a] \cdot [z-b]}$

Then there are five valid Laurent expansions:

$$\frac{1}{(z-a)(z-b)} = \frac{1}{b-a} \left(\frac{1}{z-b} - \frac{1}{z-a} \right) = \frac{1}{b-a} \left(\frac{1}{a-z} - \frac{1}{b-z} \right)$$

$$\begin{aligned} &\stackrel{(1)}{=} \frac{1}{b-a} \cdot \left(\frac{1}{a} \cdot \frac{1}{1-(z/a)} - \frac{1}{b} \cdot \frac{1}{1-(z/b)} \right) \\ &= \frac{1}{b-a} \cdot \left(\frac{1}{a} \cdot \sum_{n=0}^{\infty} \left(\frac{z}{a} \right)^n - \frac{1}{b} \sum_{n=0}^{\infty} \left(\frac{z}{b} \right)^n \right) \\ &= \frac{1}{b-a} \cdot \left(\sum_{n=0}^{\infty} \left(\frac{1}{a^{n+1}} - \frac{1}{b^{n+1}} \right) z^n \right) \quad |z| < |a| \end{aligned}$$

$$\begin{aligned} &\stackrel{(2)}{=} \frac{1}{b-a} \cdot \left(\frac{1}{z} \cdot \frac{1}{(\frac{a}{z}-1)} - \frac{1}{b} \cdot \frac{1}{1-(z/b)} \right) \\ &= \frac{1}{b-a} \cdot \left(\frac{1}{z} \cdot \sum_{n=0}^{\infty} a^n \cdot z^{-n} - \frac{1}{b} \cdot \sum_{n=0}^{\infty} \left(\frac{z}{b} \right)^n \right) \\ &= \frac{1}{b-a} \cdot \left(- \sum_{n=1}^{\infty} a^{n-1} \cdot z^{-n} - \sum_{n=0}^{\infty} \frac{1}{b^{n+1}} z^n \right) \quad |a| < |z| < |b|. \end{aligned}$$

$$\begin{aligned} &\stackrel{(3)}{=} \frac{1}{b-a} \cdot \left(\frac{1}{z} \cdot \left(\frac{1}{1-(b/z)} - \frac{1}{1-(a/z)} \right) \right) \\ &= \frac{1}{b-a} \cdot \frac{1}{z} \cdot \left[\sum_{n=0}^{\infty} \left(\frac{b}{z} \right)^n - \left(\frac{a}{z} \right)^n \right] \\ &= \frac{1}{b-a} \cdot \sum_{n=1}^{\infty} (b^{n-1} - a^{n-1}) \cdot z^{-n} \quad |z| > |b|, \end{aligned}$$

$$\begin{aligned} &\stackrel{(4)}{=} \frac{1}{b-a} \left(\frac{1}{a-b+(z-a)} - \frac{1}{z-a} \right) \\ &= \frac{1}{b-a} \left(\frac{1}{a-b} \cdot \frac{1}{1-(z-a)/(b-a)} - \frac{1}{z-a} \right) \\ &= \frac{1}{b-a} \cdot \left(\frac{1}{a-b} \cdot \sum_{n=0}^{\infty} \left(\frac{z-a}{b-a} \right)^n - \frac{1}{z-a} \right) \quad 0 < |z-a| < |a-b|. \end{aligned}$$